WORKED SOLUTION

STEP I 1987, Question 7

Pure • Trigonometric Series

STEP I 1987, Question 7

Read the question before you go on. It is in the STEP Support Programme's question database:

step.maths.org/questions/87-s1-q7

The key idea

Both series are sums of sines whose angles go up in equal steps. Multiplying a term like $\sin((2k+1)\theta)$ by $2\sin\theta$ or $2\cos\theta$ turns it, by a product-to-sum formula, into a *difference* or *sum* of two neighbouring terms, so the whole series telescopes. The only choice is which factor to use: $2\sin\theta$ suits the plain sum, and $2\cos\theta$ suits the alternating one.

Solution

Setting up

Write $\theta = \dfrac{2\pi}{23}$. The angles are $2\pi/23$, $6\pi/23$, $10\pi/23, \ldots, 42\pi/23$, increasing by $4\pi/23 = 2\theta$ each time, so they are

$$(2k+1)\theta \qquad \text{for } k = 0, 1, \ldots, 10,$$

since $21\theta = 42\pi/23$. That makes 11 terms, and in the alternating series term k has sign $(-1)^k$ (the last term, $k = 10$, is $+$, as it should be). So we want

$$S_1 = \sum_{k=0}^{10} \sin((2k+1)\theta), \qquad S_2 = \sum_{k=0}^{10} (-1)^k \sin((2k+1)\theta).$$

We shall use $22\theta = \dfrac{44\pi}{23} = 2\pi - \dfrac{2\pi}{23}$, so that

$$\cos 22\theta = \cos\theta, \qquad \sin 22\theta = -\sin\theta.$$

The first series

By the product-to-sum formula $2\sin A \sin B = \cos(A-B) - \cos(A+B)$,

$$2\sin\theta\, \sin((2k+1)\theta) = \cos(2k\theta) - \cos((2k+2)\theta).$$

Summing over $k = 0, \ldots, 10$, each $\cos(2k\theta)$ with $1 \le k \le 10$ appears once with each sign and cancels, leaving

$$2\sin\theta\, S_1 = \cos 0 - \cos 22\theta = 1 - \cos\theta.$$

Since $0 < \theta < \pi$, $\sin\theta \neq 0$ and we may divide:

$$S_1 = \frac{1-\cos\theta}{2\sin\theta} = \frac{2\sin^2(\theta/2)}{4\sin(\theta/2)\cos(\theta/2)} = \tfrac{1}{2}\tan\frac{\theta}{2},$$

using $1 - \cos\theta = 2\sin^2(\theta/2)$ and $\sin\theta = 2\sin(\theta/2)\cos(\theta/2)$, and cancelling $\sin(\theta/2) \neq 0$. With $\theta/2 = \pi/23$,

$$\boxed{\sin\tfrac{2\pi}{23} + \sin\tfrac{6\pi}{23} + \cdots + \sin\tfrac{42\pi}{23} = \tfrac{1}{2}\tan\tfrac{\pi}{23}.}$$

Ask yourself Why multiply by $2\cos\theta$ for the alternating series, rather than $2\sin\theta$ again?

With $2\sin\theta$ we get differences $\cos(2k\theta) - \cos((2k+2)\theta)$, and the alternating signs would make neighbouring terms *add* rather than cancel. With $2\cos\theta$ we get *sums* $\sin((2k+2)\theta) + \sin(2k\theta)$, and now it is the alternating sign that produces the cancellation.

The second series

By $2\cos A\sin B = \sin(B+A) + \sin(B-A)$,

$$2\cos\theta\,\sin((2k+1)\theta) = \sin((2k+2)\theta) + \sin(2k\theta).$$

Multiply by $(-1)^k$ and sum. For $1 \le k \le 10$, $\sin(2k\theta)$ appears from term $k-1$ with sign $(-1)^{k-1}$ and from term k with sign $(-1)^k$, so it cancels. What survives is $\sin 0$ from $k = 0$ and $(-1)^{10}\sin 22\theta$ from $k = 10$:

$$2\cos\theta\, S_2 = \sin 22\theta = -\sin\theta.$$

Since $0 < \theta < \pi/2$, $\cos\theta \neq 0$, so

$$S_2 = -\frac{\sin\theta}{2\cos\theta} = -\tfrac{1}{2}\tan\theta,$$

that is,

$$\boxed{\sin\tfrac{2\pi}{23} - \sin\tfrac{6\pi}{23} + \cdots - \sin\tfrac{38\pi}{23} + \sin\tfrac{42\pi}{23} = -\tfrac{1}{2}\tan\tfrac{2\pi}{23}.}$$

If you prefer a positive-looking form, this is also $\frac{1}{2}\tan\frac{21\pi}{23}$, since $\tan(\pi - x) = -\tan x$.

Checks

- Numerically, $\frac{1}{2}\tan(\pi/23) \approx 0.068723$ and $-\frac{1}{2}\tan(2\pi/23) \approx -0.140093$; adding up the eleven terms directly gives the same values to 15 significant figures.

- The negative sign of the second answer is believable: the partial sums swing between about ± 0.5, and the last term added, $\sin(42\pi/23) \approx -0.27$, is itself negative.
- The same method gives the general formulas $\sum_{k=0}^{n-1} \sin((2k+1)\theta) = \dfrac{\sin^2 n\theta}{\sin\theta}$ and $\sum_{k=0}^{n-1} (-1)^k \sin((2k+1)\theta) = (-1)^{n-1} \dfrac{\sin 2n\theta}{2\cos\theta}$; try $n = 1$, where both reduce to $\sin\theta$.

Common slips

- Miscounting the terms: the angles step by $4\pi/23$, not $2\pi/23$, so there are 11 terms, not 21.
- Expecting both answers to be positive. Angles beyond π have negative sines, and $\sin(44\pi/23) = -\sin(2\pi/23)$ is where the minus sign comes from.

Beyond STEP

These sums are the imaginary parts of geometric series: $S_1 = \operatorname{Im} \sum \mathrm{e}^{\mathrm{i}(2k+1)\theta}$, and S_2 is the same with ratio $-\mathrm{e}^{2\mathrm{i}\theta}$, which gives a second route to both answers. Sums of this kind, such as the Dirichlet kernel, are the starting point of Fourier analysis, where a function is built from sines and cosines. Needham's *Visual Complex Analysis* shows why such sums behave as they do, using pictures of rotating vectors.

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